Stanley Li

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Education

University of California, Berkeley Aug. 2024 – May 2028
B.A. in Computer Science and Applied Math *Berkeley, CA*

- GPA: 3.9/4.0
- Honors & Awards: USA Physics Olympiad Silver Medal, AIME Qualifier (2x)
- Coursework: Graduate-level Theoretical Statistics, Data Structures and Algorithms, Computer Architecture, Probability & Random Processes, Linear Algebra, Statistical Physics, Discrete Math, Real Analysis

Experience

Susquehanna International Group June 2027 – August 2027
Quantitative Trader Intern *Bala Cynwyd, PA*

- Offer received for Summer 2027

MATS (ML Alignment & Theory Scholars) June 2026 – Present
Research Fellow *Berkeley, CA*

- Built finetuning pipelines for model organisms, including RL via GRPO and CoT-masked SFT to induce secret keeping and denial. Verified by RL'ing Gemma-3-4B from 54% to 65% accuracy on multiplication dataset
- Jailbreaking secrets using activation fuzzing and interpretability methods including steering vectors

METR (Model Evaluation & Threat Research) May 2026
Research Contractor *Berkeley, CA*

- Testing activation fuzzing methods for secret elicitation, including MELBO-trained steering vectors, successfully jailbreaking Gemma-4-31B and Gemma-4-E4B

Swoop Dec. 2025 – Jan. 2026
Software Engineer Intern - Backend *Los Angeles, CA*

- Designed REST API configuration service using Django/DRF with Redis caching, enabling localization and app content updates for 10,000+ users without redeployment
- Designed search service using Elasticsearch with Docker, allowing for localized search and autocomplete

Lawrence Berkeley National Lab Sep. 2025 – Present
Undergraduate Researcher - Perlmutter group *Berkeley, CA*

- Using NERSC HPC system to run GIGALens model on multiple GPU nodes, modeling gravitational lensing
- Using JAX to perform experiments including testing SVI numerical stability under different priors/optimizers

Projects

CubeGPT - Transformer-based Rubik's Cube Solver (GitHub) | *Python, PyTorch* June 2025

- Built Rubik's Cube solver using beam search and transformer model evaluation, achieving 100% solve rate with average solve length 19.0 moves (optimal bound 17.7)
- Trained 30+ transformers on 300M+ generated data samples, iterating on architecture and hyperparameters
- Implemented MoE with routing bias (Deepseek V3 style), gated MLP, and canonicalization, achieving 7x sample efficiency over out-of-the-box transformer. Reached 5% lower MSE with 2x less training.

ML Library from Scratch (GitHub) | *Python, NumPy, SciPy, Matplotlib* Feb. 2025

- Built ML library from scratch supporting CNNs, ResNets, and MLPs for training classifiers, autoencoders, and diffusion models with Keras-inspired declarative API
- Implemented standard data augmentations, regularization, and optimizers including Adam, training an MNIST model to 98%+ accuracy and enabling interpretable feature visualization

Skills

Languages: Python, C/C++, Java, SQL
Libraries: PyTorch, TensorFlow/Keras, NumPy, SciPy, pandas, Matplotlib, Django/DRF, OpenGL, GLFW
Developer Tools: Git, GitHub, Docker, Visual Studio, VS Code, IntelliJ, Eclipse
Interests: Competitive Typing (230+ WPM), History, Creative Writing, Sci-Fi